Evaluation of catalysts for the hydrogenation of CO_2 to ethanol

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Global warming is one of the greatest challenges of our time. CO_2 is a major greenhouse gas, and its capture or conversion plays a substantial role against global warming. Among potential products where CO_2 serves as the carbon source, ethanol (EtOH) is on high demand due to its use as fuel additive and chemical building block. Research shows however, that the number of studies investigating hydrogenation of CO to EtOH exceeds that of CO_2 to EtOH (Figure 1a and 1b respectively). Therefore, the goal of this study is to obtain data of catalysts for CO conversion from literature under identical conditions to evaluate their performance in CO_2 conversion. Furthermore, the study will explore how CO_2 affects the catalytic performance in a CO_2 feed and simulate the use of mixed feeds, such as recycled or reverse water gas shift product feeds. In CO conversion, the RhMnbased[1] systems show the highest selectivity's between 20 and 40 %. CuZn-based[3,4] (modified methanol) and CoCu-based[2] (modified Fischer Tropsch) catalysts exhibit in most cases selectivity's below 20 % but can enable higher conversions. CoCu-based catalysts are promising combinations, as Co is an element that supports the formation of Fischer-Tropsch products and Cu could provide the active sites for the formation of oxygenates. However, EtOH yields are mostly limited to 10 %. The performance decreases further when converting CO_2 (Figure 1b).

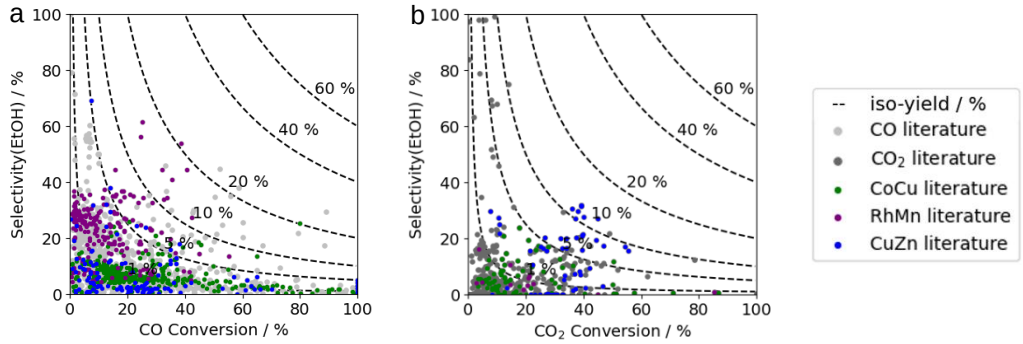

Figure 1. EtOH selectivity for a) CO conversion and b) CO_2 conversion grouped by composition.

Based on literature research, the lead candidate for each of the three mentioned catalyst groups were selected. In addition, a $CuAlZnO_x$ catalyst was modified with Cs as an alkali metal as this leads to an enhanced selectivity to higher alcohols.[4] In order to determine the effect of CO_2 in the feed a CO_2/CO_x variation was conducted. $CoCuMnO_x$ exhibits an increasing selectivity towards methane and a decreasing selectivity for lower alkanes and C_{2+} oxygenates. $RhMnIrLi/SiO_2$ shows a high selectivity of 37 % for EtOH and additionally a selectivity of 15 % towards higher alcohols and C_{2+} oxygenates. While in the case of a mix feed the selectivity remains stable since mainly the CO in the feed is converted. the EtOH selectivity is reduced and methanation increases under pure CO_2 feed. For both $CuAlZnO_x$ catalysts, methanol can be recognized as the main product for the CO and the mixed feed. While pure CO feed also has a low selectivity for alcohols and C_{2+} alkanes, it decreases for the mixed feed. In the case of a pure CO_2 feed, CO is the main product, while the selectivity to methanol amounts 15 %. In general, the Cs-promoted catalyst shows no significant differences in the product distribution.

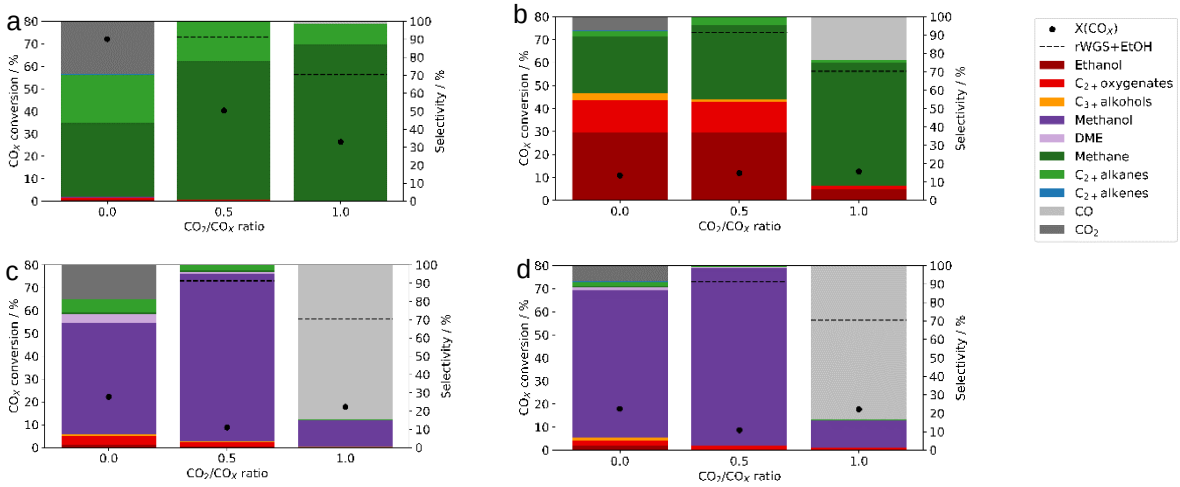

Figure 2. Product-selectivity's for a CO (left), a mixed (middle) and a CO_2 (right) feed at T = 270 °C, WHSV= 3000 mL/g*h, H_2/CO_x = 3 for a) $CoCuMnO_x$, b) $RhMnIrLi/SiO_2$ c) $CuAlZnO_x$ and d) Cs-$CuAlZnO_x$.

Apart from $RhMnIrLi/SiO_2$, the other catalytic systems are not suitable for EtOH formation in CO_2. These results show, that a transfer to CO_2 conversion is not facile without further modifications to the catalyst system. Methanation remains the main bottleneck, reducing the selectivity to EtOH. The catalysts will have to be further developed especially with regard to the suppression of the undesired methane.

References

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